Final Project

The objective of this project is to give students experience with real data analysis. Each student will collect and analyze a data set of their choosing using one or more of the statistical methods covered in this course. More advanced techniques are permitted if the student considers them necessary.
The data set may reflect a personal or academic interest, but it must be real, must suit one of the analyses studied in this course, and must not have been analyzed by the student previously or in this course. The sample size should be relatively large. Data from all various sources are acceptable. Here are some recommended data archives to explore:

- Kaggle
- UC Irvine Machine Learning Repository
- Google Dataset Search
- Data.gov
- OpenML

The project counts for 20% of the course grade and consists of two parts.

Part I: Project Report (15%)
The project report is due in Canvas on **December 6** by 11:59PM. It should summarize the analysis concisely and run no longer than five pages, excluding software output, graphs, and other supporting material. The report should address the following:

- The topic selected and the reason for the student's interest in it
- A brief description of the series analyzed
- The stationarity or nonstationarity of the series
- The model selected and the model building procedure
- Results of any relevant hypothesis tests or confidence intervals
- Model adequacy checking, including remedial methods for any violations observed
- Overall conclusions about the data based on the analysis

Part II: Project Presentation (5%)
Presentations will be given during class time on **December 3**. Each student will present the project with slides in no more than 15 minutes. Presentations will be evaluated on:

- Clarity of delivery
- Knowledge of the topic and related issues
- Organization of the presented material
- Confidence of the speaker